

DILSHAN CHAMILA

AI / ML Intern

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SUMMARY

Software Engineering undergraduate specializing in Machine Learning, Computer Vision, and Predictive Modeling. Hands-on experience building and deploying end-to-end AI/ML systems using Python, TensorFlow, OpenCV, and Scikit-learn. Proven ability to train, evaluate, and optimize models across classification, regression, and deep learning tasks, with full-stack integration skills to deliver production-ready AI-powered applications.

EDUCATION

BSc (Hons) Software Engineering
Sabaragamuwa University of Sri Lanka

2023 – Present (Expected 2027)
Current GPA: 3.32 / 4.00

SKILLS & TECHNOLOGIES

- **AI / ML:** Machine Learning, Deep Learning, Computer Vision, TensorFlow, OpenCV
- **Data Science:** Scikit-learn, Pandas, NumPy, Matplotlib
- **Languages:** Python, JavaScript, TypeScript, HTML5, CSS3
- **Backend & API:** FastAPI, Flask, Node.js, Express.js, REST APIs
- **Frontend:** React.js, Next.js, Tailwind CSS, React Native (Expo)
- **Databases:** PostgreSQL, MongoDB, MySQL
- **Version Control:** Git, GitHub
- **Cloud & Tools:** Google Cloud, Firebase, Postman, VS Code, IntelliJ IDEA, Jupyter Notebook
- **Project Management:** Jira, Agile

PROJECTS

ATM Guard AI: Real-Time Surveillance System *(Group Project)*

[Live Demo](#) [GitHub](#)

- Led a team of 2 members and implemented AI/ML models using TensorFlow and OpenCV, achieving ~80% accuracy; developed a backend processing system for continuous data handling; created a monitoring dashboard for user interaction.
- Designed an AI-powered surveillance system for ATM security, training on 1,000+ images with data augmentation, achieving ~80% accuracy in threat detection.
- Engineered 4+ Express.js API endpoints to bridge the Python ML pipeline with the React.js dashboard, delivering real-time threat alerts with under 1-second response latency during live surveillance.
- **Tech Stack:** TensorFlow, OpenCV, Python, Node.js, Express.js, React.js, MongoDB

Student Performance Predictor *(ML Classification Model)*

[GitHub](#)

- Built a binary classification model using Scikit-learn to predict student pass/fail outcomes, achieving ~92% accuracy across 2+ academic feature categories through iterative hyperparameter tuning.
- Performed end-to-end data preprocessing and feature engineering handling missing values, encoding 5+ categorical variables, and normalizing numerical features to optimize model input quality.
- Evaluated model performance using accuracy, precision, and confusion matrix metrics, achieving ~0.5 second prediction time during local testing.
- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, FastAPI

LapPredict – Laptop Price Prediction System

[GitHub](#)

- Developed a Random Forest regression model to estimate laptop prices based on 6+ hardware specifications (brand, RAM, processor, storage, GPU, display), achieving ~75% prediction accuracy on a 1,000+ record dataset.
- Cleaned and preprocessed 1,000+ raw records using Pandas and NumPy, resolving inconsistencies, encoding categorical variables, and engineering features to improve regression model performance.

- Created a Flask-based web application for laptop price estimation, delivering predictions in ~1 second through a responsive HTML/CSS interface.
- **Tech Stack:** *Python, Scikit-learn, Pandas, NumPy, Flask, HTML/CSS, Jupyter Notebook*

InterviewGenie: AI-Powered Interview Assistant *(Group Project)*

[Live Demo](#) [GitHub](#)

- Engineered an AI-powered mock interview platform using Gemini AI (~1K token-based requests) to generate questions and deliver instant feedback across multiple sessions.
- Built a full-stack application using Next.js and Node.js, supporting 10+ interview sessions; integrated Gemini AI for automated question generation; enforced secure authentication for multi-user access.
- Configured Next.js server-side rendering (SSR) for 3+ core pages, reducing initial load latency and improving responsiveness for concurrent users across varying network conditions.
- **Tech Stack:** *Next.js, React, Node.js, PostgreSQL, Drizzle ORM, Tailwind CSS, Gemini AI*

CERTIFICATIONS

- **Introduction to Generative AI** | Google Cloud Skills Boost – 03/2025 [🔗](#)
- **Web Development** | Sololearn – 03/2025 [🔗](#)
- **Python for Everyone** | Udemy – 01/2026 [🔗](#)
- **Python for AI & Machine Learning** | Udemy – 01/2026 [🔗](#)

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Announcing Crew Member | TECHTALK v1.0 | Society of Computer Science, SUSL [🔗](#)
- Knowledge Team Member | Hope 1.0 | IEEE WIE Student Branch Affinity Group, SUSL [🔗](#)
- Announcer | Announcing Circle, Sabaragamuwa University of Sri Lanka [🔗](#)

AWARDS & RECOGNITION

- Semi-Finalist | Hackventure Competition 1.0 – Team-based national hackathon [🔗](#)